



NICHOLAS LAURENSIUS HALIM

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PROFILE

Undergraduate Computer Engineering student specializing in Embedded Systems, IoT, and Network Infrastructure. Experienced in end-to-end hardware development, from PCB layout and FPGA logic (VHDL) to RTOS firmware and cloud synchronization. Adept at transforming complex engineering concepts into functional, reliable technological prototypes.

EDUCATION

Bina Nusantara University (Binus University) | *Bachelor of Computer Engineering* 2024 - Present

- IPK: 3.47
- Relevant Coursework: Computer System Architecture, Computer Networks, Microcontroller, Internet of Things, Digital Systems.

TECHNICAL PROJECT & SYSTEM DEVELOPMENT EXPERIENCE

IoT System Developer | Smart Parcel Box (Square) Feb 2026 - Jun 2026

- Engineered an IoT-based secure parcel locker using ESP32-S3 and FreeRTOS to orchestrate network tasks, physical interfaces, state machines, and media processing concurrently without blocking.
- Integrated a VL53L0X Time-of-Flight sensor for capacity detection, an OV2640 camera for MJPEG video/photo evidence, and a solenoid lock validated via a 3x4 matrix keypad.
- Developed an offline-to-online failsafe architecture that buffers sensor data and media on a MicroSD card during network outages, and automatically synchronizes with Firebase Realtime Database and Cloud Storage upon connection restoration.

Linux System Administrator | Pi-hole Network-Wide Ad Blocker Feb 2026 - Jun 2026

- Repurposed an unused Android Set-Top Box (B860H) into a local DNS sinkhole using Armbian Linux and Pi-hole to block network-wide advertisements and trackers.
- Managed the Linux server remotely via SSH, configured router DHCP reservations, and automated daily system maintenance tasks using cron jobs.

Network Engineer | Multi-Building Campus LAN Infrastructure (CNIS) Feb 2026 - Jun 2026

- Architected and simulated a 3-building campus network infrastructure utilizing Cisco Packet Tracer, implementing a Hierarchical Star Topology.
- Configured OSPF Multi-Area routing for optimal inter-building communication and deployed VLAN segmentation with Inter-VLAN routing via Multilayer Switches.
- Implemented defense-in-depth network security mechanisms, including Extended ACLs, DHCP Snooping, and Port Security, to mitigate unauthorized access and rogue DHCP servers.

Hardware & Firmware Developer | Smart Mini Vending Machine Sep 2025 - Jan 2026

- Built a fully functional mini vending machine prototype for distributing electronic components, powered by an ATmega328P (Arduino Nano) and programmed in pure C (AVR-GCC).
- Designed a robust Finite State Machine (FSM) firmware featuring hardware interrupts for coin detection (optocoupler) and an automated recovery mechanism utilizing a Watchdog Timer.
- Authored custom PCB schematics and layouts via EasyEDA, and fabricated the mechanical components and acrylic enclosure using 3D printing and laser cutting.

IoT & Edge AI Developer | Smart Sensor Hub & Vision Classifier Mar 2026 - Apr 2026

- Developed a real-time sensor hub using FreeRTOS and authenticated MQTT telemetry to publish environmental data.
- Deployed a quantized TinyML vision classifier via Edge Impulse directly onto a microcontroller for local, non-contact object sorting with >90% accuracy.

ADDITIONAL EXPERIENCE & VOLUNTEERING

E-commerce Operations Assistant | Family Business (Shopee) 2024 - Present

- Managed physical and digital inventory synchronization, alongside product photography and descriptions.

Device Preparation Technician | Chromebook Digital Library (PusGita) Mar 2026

- Performed physical inspections and functional testing on 5 Chromebook clients and 1 Raspberry Pi server to ensure operational readiness for a remote digital library system.
- Managed system setups including network credentials, login systems, and SD card cloning for the server deployment.

Exhibitor & Event Logistics | Computer Engineering All For One (CEAFO) 2023 - 2025

- Coordinated equipment requirements for organizational events and successfully demonstrated technical prototypes to non-technical audiences.

EXPERTISE

- Microcontrollers & Hardware: ESP32-S3, Arduino/ATmega328P, FPGA (Nexys A7-100T), PCB Design (EasyEDA), I2C/SPI Protocols.
- Programming & Firmware: C/C++, VHDL, FreeRTOS, Finite State Machine (FSM), Hardware Interrupt Management.
- Networking, Cloud & OS: Cisco IOS, OSPF Multi-Area, VLAN, Firebase (RTDB & Storage), MQTT, Armbian Linux.
- Tools & Prototyping: Git/GitHub, Cisco Packet Tracer, Vivado, 3D Modeling (Tinkercad), 3D Printing (FDM).